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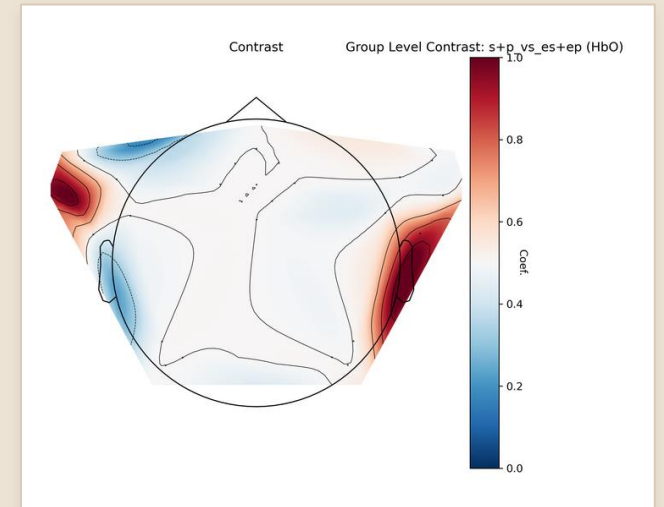
RESEARCH CASE STUDY · MASTER'S INTERNSHIP · 2025

How do we know if a driverless car will let us cross the street?

A behavioral and brain-imaging (fNIRS) study of how pedestrians infer the yielding intention of autonomous vehicles, and whether they use the same mechanisms as when reading human minds.

MASTER'S THESIS

Intention Inference Toward Autonomous Vehicles: Behavioral and fNIRS Evidence for Shared and Distinct Mechanisms with Human Mentalizing



Study at a glance

The question. Pedestrians cannot make eye contact with a driverless car. I studied how people infer whether an autonomous vehicle (AV) intends to yield, and whether they use Theory of Mind (ToM), the ability we use to read other people's intentions.

The answer. People reliably read AV intent from changes in speed. But ToM played no detectable role: individual ToM ability did not predict AV judgments, and the brain region engaged by the ToM task was not engaged by the AV task. This points to a simpler, non-mentalizing mechanism.

Context	Master's research internship, EMC Lab (Laboratoire d'Étude des Mécanismes Cognitifs), Université Lumière Lyon 2
Supervisors	Jordan Navarro, Emmanuelle Reynaud
My role	Ran all experimental sessions, analysed the behavioral and fNIRS data, and wrote the research report. The study was designed by my supervisors.
Duration	4 months, September to December 2025
Participants	44 healthy young adults (30 women, mean age 24.5)
Methods	fNIRS brain imaging, Theory of Mind task, continuous joystick ratings, mixed-effects models, GLM
Tools	Python (MNE-NIRS), R (lme4), Cortiview

Without a driver, pedestrians must read the car's motion

WITH A DRIVER

Motion + driver cues

Pedestrians read the car's speed and distance, plus the driver's eye contact and gestures, to judge whether it will yield (Sucha et al., 2017).



WITHOUT A DRIVER

Motion only

No driver to look at. Even when the car shows external displays (eHMI), pedestrians still decide mainly from speed and distance (Clamann et al., 2017; Dey et al., 2017; Rothenbücher et al., 2016).



MOTION CARRIES INTENT

Speed changes matter

Not only how fast the car goes, but how its speed changes: abrupt acceleration is perceived as erratic and harms crossing decisions (Zimmermann & Wettach, 2017).

THE OPEN QUESTION

So motion is the AV's real interface. But how do pedestrians decode it?

Theory explains how we read people, not machines

1

Intentions shape movement

Placing an object carefully, rather than throwing it, lengthens the deceleration phase (Marteniuk et al., 1987). Social and individual grasps differ in finger aperture (Becchio et al., 2008).

2

Humans read these kinematics

Observers use subtle timing differences to infer social intentions (Lewkowicz et al., 2015). This is Theory of Mind (ToM): the ability to infer others' thoughts and intentions to predict their behavior (Premack & Woodruff, 1978).

3

The brain treats machines differently

The mentalizing network (TPJ, mPFC, MTG) is less active for robots than for humans (Hortensius & Cross, 2018). The TPJ responds to a robot only when people believe a human controls it (Özdem et al., 2017).

THE RESEARCH GAP

No study had directly tested whether inferring an AV's intention relies on the same mechanisms as ToM, at both the behavioral and the neural level.

TPJ: temporo-parietal junction. mPFC: medial prefrontal cortex. MTG: middle temporal gyrus.

What the study set out to answer

Does inferring an autonomous vehicle's intention rely on the same mechanisms as reading a human mind?

H1 Sensitivity to speed profile

BEHAVIORAL PREDICTION

Speed changes (acceleration or deceleration) produce stronger intent inference than constant speed.

NEURAL PREDICTION

These differences appear as distinct patterns of brain activation.

H2 Partial overlap with Theory of Mind

BEHAVIORAL PREDICTION

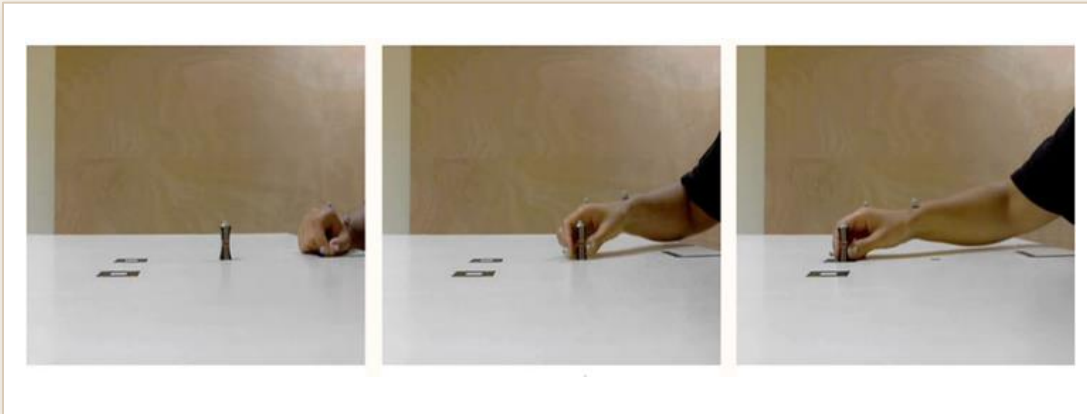
People with higher ToM sensitivity produce more coherent AV intent judgments.

NEURAL PREDICTION

The AV task recruits some ToM regions, but more weakly, because an AV is less socially salient than a human.

Two tasks and two continuous brain recordings

44 healthy young adults (45 recruited, 1 excluded) · 30 women · mean age 24.5 (SD 3.8) · within-subject design



TASK 1 · THEORY OF MIND (TOM)

Reading human intent

41 trials. Participants watched reach-to-grasp videos and judged whether the action had a social intent (placing an object for a partner) or an individual intent (for personal use), pressing a key as fast and accurately as possible. Stimuli adapted from Lewkowicz et al. (2015).

Social vs individual intent · keyboard response



TASK 2 · SIMPAD

Reading AV intent

24 trials (6 per condition). From a roadside pedestrian view, participants watched a simulated AV approach and rated its yielding intent continuously with a joystick: forward means it will yield, backward means it will not, and deflection size means certainty.

30 km/h constant

50 km/h constant

SlowDown

SpeedUp

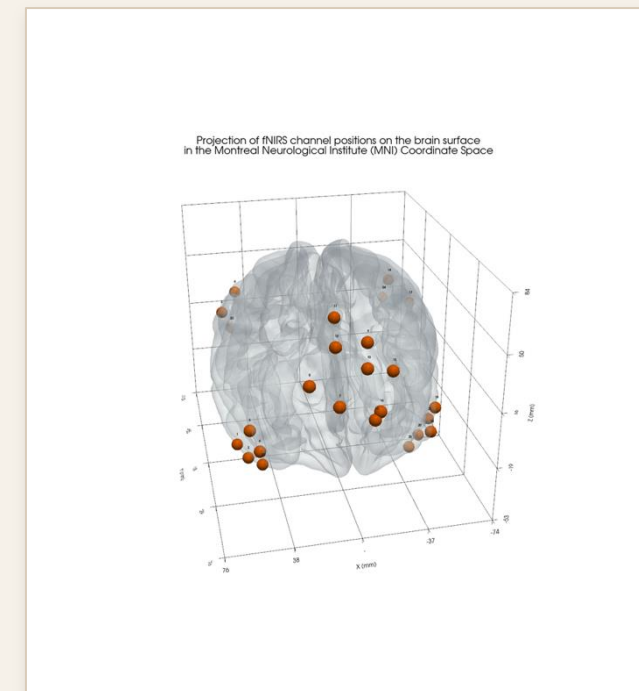
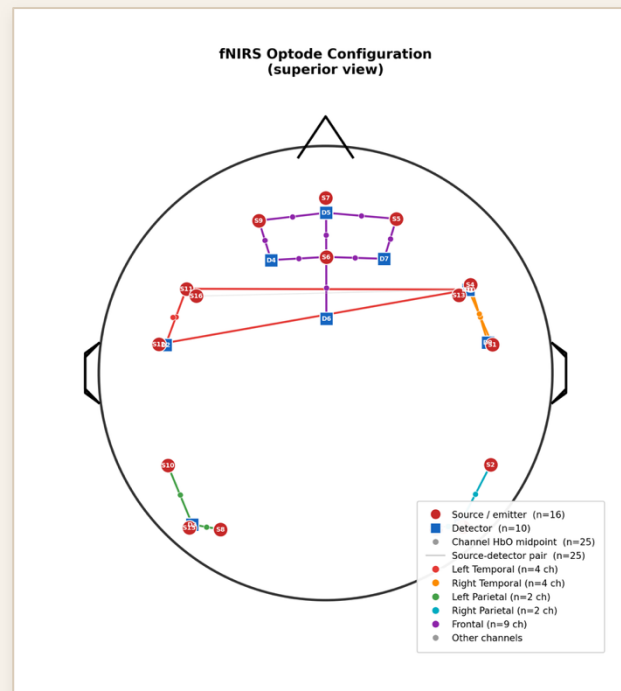
Measuring brain activity with light

What is fNIRS?

Functional near-infrared spectroscopy is a wearable brain-imaging method. Light sources on a cap send near-infrared light through the scalp, and detectors measure how much returns. This reflects changes in blood oxygenation in the outer layer of the cortex, an indirect marker of neural activity.

SETUP

- Photon Cap C20 fNIRS system, recorded with Cortiview
- 25 channels grouped into 5 regions of interest (ROIs): frontal (9), left and right temporal (4 each), left and right parietal (2 each)



Optode layout seen from above (left) and channel positions projected onto a standard brain in MNI space (right).

From raw signals to statistical tests

BEHAVIORAL DATA

ToM sensitivity

d' (d-prime) from signal detection theory: each participant's ability to tell social from individual intent, used as a continuous measure of mind-reading ability.

H1 · Speed effect

Linear mixed-effects model (LMM): mean absolute joystick displacement ~ speed condition, random intercept per participant; Tukey-adjusted post-hoc tests.

H2 · Cross-task link

Alignment index: each participant's joystick amplitudes correlated with the predicted gradient (SlowDown > 30 > 50 km/h > SpeedUp), then regressed on d' .

FNIRS DATA

I · Preprocessing (Python, MNE-NIRS)

Wavelet motion correction, 0.01–0.1 Hz band-pass filter, modified Beer–Lambert law to obtain oxygenated hemoglobin (HbO) per channel.

II · First level (per channel)

General linear model (GLM): condition regressors and a joystick parametric modulator, convolved with a canonical hemodynamic response function, give a β per condition.

III · Group level (per ROI, R/lme4)

H1: one-sample t-tests on the joystick β , FDR-corrected. H2: ToM intent vs control contrast; each AV condition vs the inter-trial baseline.

Four analytic choices

01 Why d' rather than accuracy?

d' separates true sensitivity from response bias, such as a tendency to answer “social” more often. This gives a cleaner continuous measure of individual mind-reading ability.

02 Why absolute joystick displacement?

Taking the absolute value measures how strong and certain a judgment is, regardless of whether the participant inferred correctly the car to yield or not.

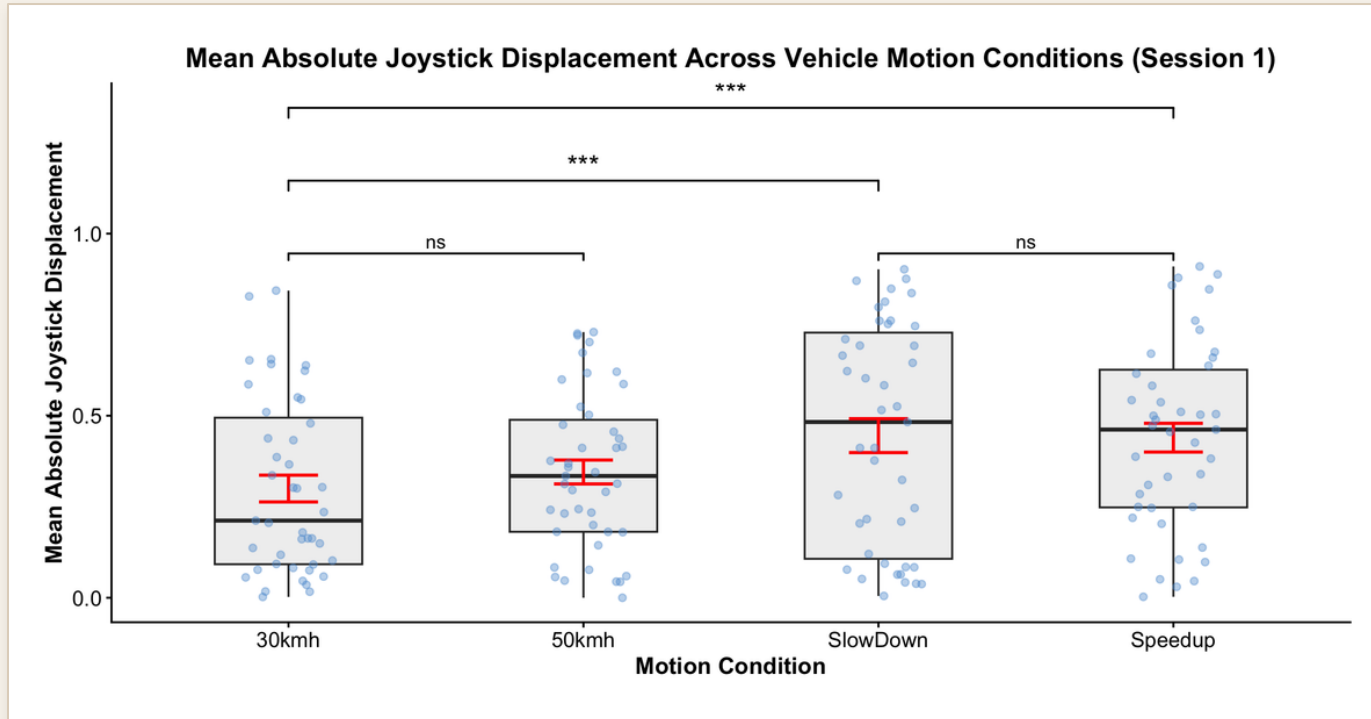
03 Why a mixed-effects model?

Every participant saw all four conditions. A random intercept models each person's own baseline, so condition effects are not confused with individual rating habits.

04 Why HbO and FDR correction?

HbO has a better signal-to-noise ratio than deoxygenated hemoglobin. FDR (false discovery rate) correction limits false positives across the many ROI × condition tests.

A speed change, not its direction, drives intent



Mean absolute joystick displacement by condition. *** $p < .001$; ns, not significant.

All conditions vs zero $p < .0001$

SlowDown vs 30 km/h $\Delta M = 0.145, p < .001$

SpeedUp vs 30 km/h $\Delta M = 0.140, p < .001$

SpeedUp vs SlowDown $p = .999$

WHAT IT MEANS

The presence of a speed change, not whether the car speeds up or slows down, produced stronger and more confident intent judgments. Because the measure is absolute, it reflects judgment strength, not the predicted yielding direction.

Judgment strength was not reflected in cortical activity

pFDR $\approx$.81

for every ROI and condition when testing whether HbO activation scaled with joystick intensity.

RESULT

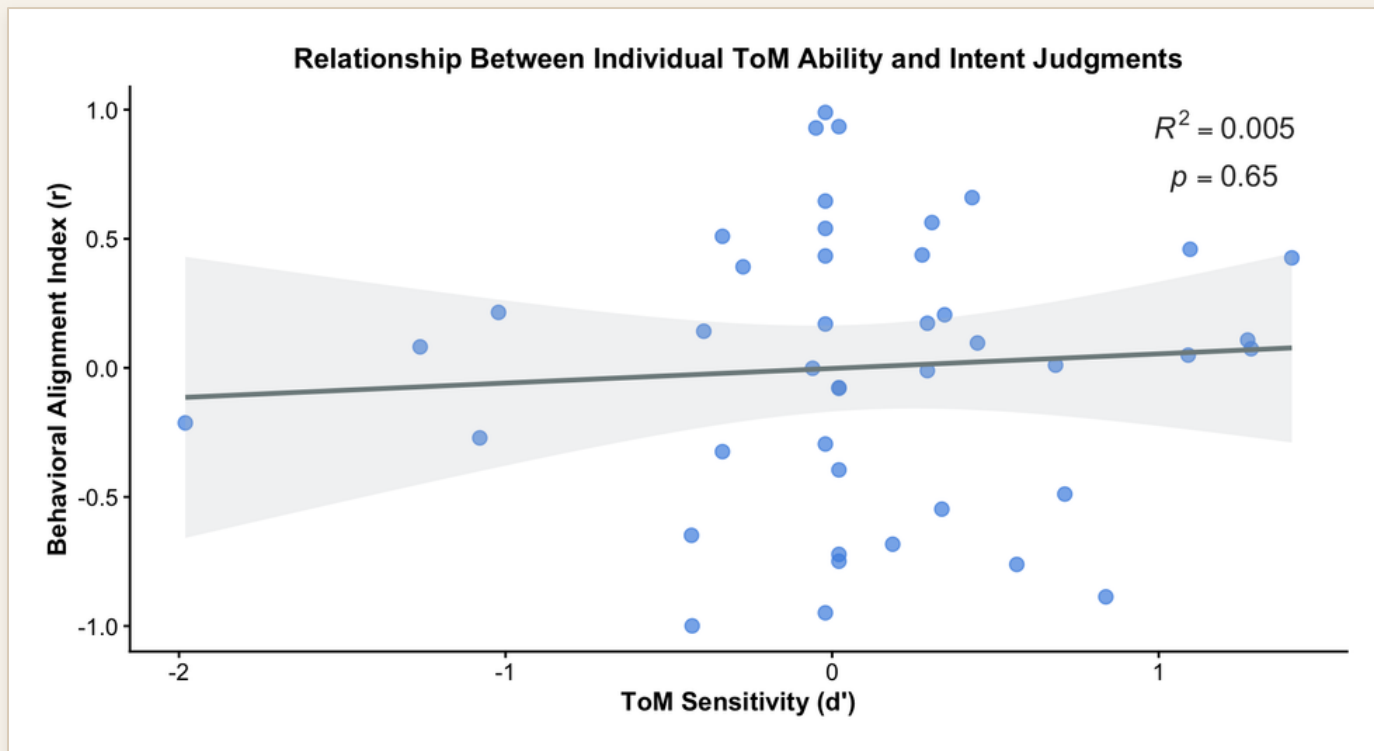
Across all five ROIs and four conditions, the slope linking cortical activation to joystick intensity never differed reliably from zero after FDR correction.

INTERPRETATION

Within this paradigm and the measured regions, how strongly people inferred intent did not linearly modulate cortical hemodynamics. A null result like this cannot show that no relation exists; it shows that none was detected.

H1 verdict: partially supported. Behavior was sensitive to the speed profile, but brain activity differed only under SpeedUp (Finding 04) and did not scale with judgment strength.

Mind-reading ability did not predict AV judgments



Each dot is one participant: ToM sensitivity (d') against behavioral alignment with the predicted yielding gradient.

Regression coefficient $\beta = 0.057$

Significance $p = .650$

Explained variance $R^2 = .005$

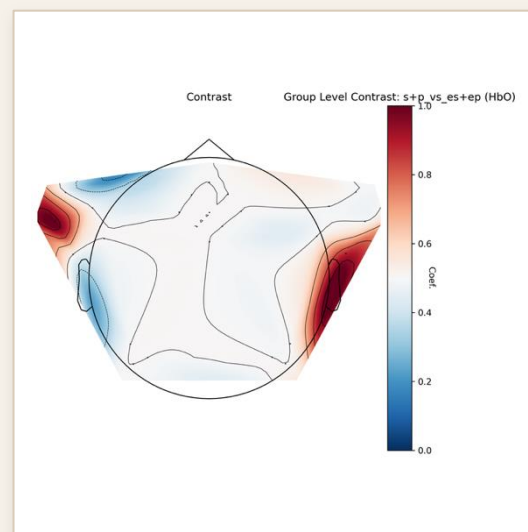
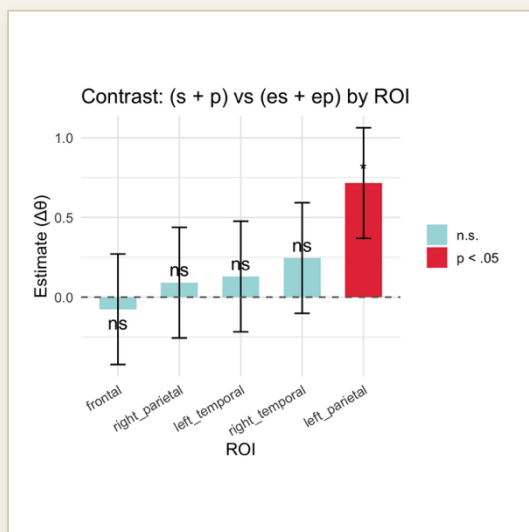
WHAT IT MEANS

People who were better at reading human intentions were not better at producing gradient-consistent judgments about the AV.

Caveat: most d' scores were near chance, which weakens this test (see Limitations).

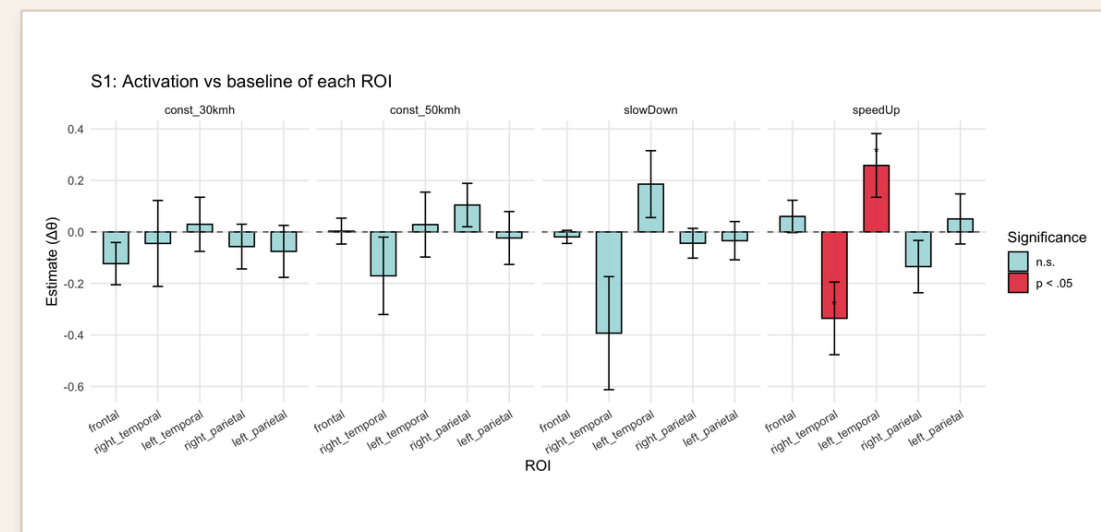
The two tasks engaged different brain regions

TOM TASK · INTENT VS CONTROL



Only the left parietal ROI was significant ($\Delta\theta = 0.716$, $p = .040$); all other ROIs $ps > .480$.

AV TASK · EACH CONDITION VS BASELINE



Only SpeedUp showed effects: left temporal $\uparrow$ ($\Delta\theta = 0.258$, $p = .021$), right temporal $\downarrow$ ($\Delta\theta = -0.335$, $p = .036$); all others $ps > .084$.

No overlap. The region recruited by mind-reading (left parietal) showed no engagement during the AV task. $\Delta\theta$ is the group-level estimate of the change in HbO for a contrast.

A convergent dissociation, and what may explain it

HYPOTHESIS SCORECARD

H1 · behavioral Speed change strengthens judgments	Supported
H1 · neural Distinct activation; scaling with strength	Partial
H2 · behavioral ToM sensitivity predicts AV judgments	Not supported
H2 · neural Overlap with ToM regions	Not supported

TWO WAYS TO INFER INTENT

Mentalizing (ToM)

Infers hidden mental states, such as beliefs and intentions, behind an agent's actions.

Teleological stance

Attributes goals from observable motion, assuming the agent acts rationally, without inferring a mind (Gergely & Csibra, 2003).

INTERPRETATION

The results are consistent with pedestrians using the teleological stance: under time pressure, they track speed changes to predict what the AV will do, rather than attributing a mind to it. This is an interpretation, not something the study tested directly.

What this could mean for AV motion design

Design hypotheses derived from a lab study with young adults and simulated videos. They need validation in realistic traffic before being applied.

01

Make yielding visible through speed change

Constant speed, especially at 30 km/h, produced weaker intent judgments than speed changes. A clear, early deceleration is likely the most legible way for an AV to signal that it will yield.

02

Treat acceleration as a strong signal

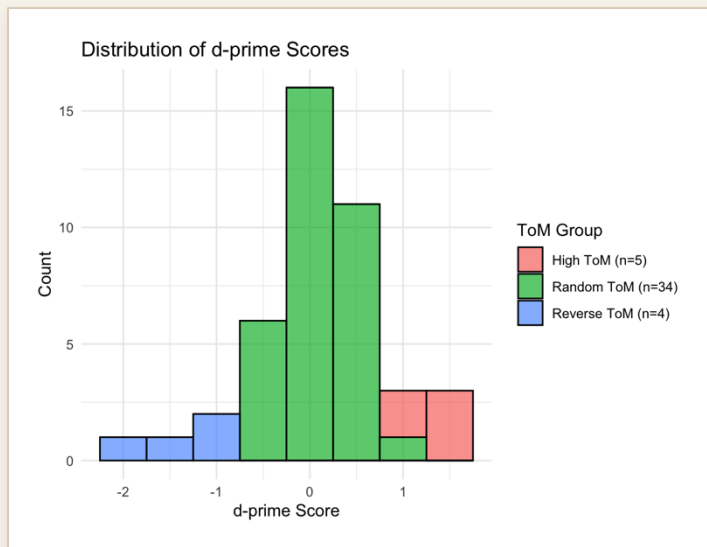
SpeedUp produced judgments as strong as SlowDown and was the only condition with distinct brain responses, possibly reflecting heightened attention. Accelerations near pedestrians should be designed deliberately.

03

Do not assume people read an AV's mind

If pedestrians track motion rather than attribute a mind, legible kinematics matter more than social or human-like cues. eHMIs should reinforce the vehicle's motion, not replace it.

Three limitations, and how to address them



Distribution of ToM d' scores: most participants fell in the near-chance group (green).

FUTURE RESEARCH

Examine the neural bases of the teleological stance directly, use more ecologically valid traffic scenarios, and test whether the ToM dissociation is amplified or abolished in neurodiverse populations.

LIMITATION

NEXT STEP

- 1 ToM scores near chance**
Range restriction weakens correlations, so a true dissociation cannot be told apart from a task that was not sensitive enough. → Use a ToM task calibrated to reveal individual differences.
- 2 No resting-state baseline**
Absolute hemodynamic responses in the AV task cannot be interpreted. → Add a dedicated baseline block to the protocol.
- 3 Averaged joystick signal**
Mean displacement collapses the time course of each judgment. → Analyse time-resolved trajectories to see whether intent builds up or appears at kinematic events.